Siddhartha Lal
29/2/24

PH1201 Electricity & Magnetism — **Mid-semester Examination**

Instructor: Dr. Siddhartha Lal — **Spring Semester, 2024**

Instructions: Answer all questions for full marks. Put your name and roll number on both Question paper and Answer script, and hand both in. $e = -1.6\times10^{-19}$C, $1/4\pi\epsilon_0 = 9\times10^9 Nm^2/C^2$

Date: February 29, 2024 **Full marks: 20** **Time: 1.5 hours**

Name: **Roll Number:**

Q1. Conservative Electric Field (or not)? [2 marks]

Check whether the following potential function leads to a conservative electric field or not:

$$V(x, y, z) = xy^2 + x^2y + z^3 \ .$$

Q2. Electric field of a uniformly charged ring. [6 marks]

(i) Show that the field along the axis of a uniformly charged ring of radius R is maximum at a distance $R/\sqrt{2}$ from the plane containing the loop.
(ii) What happens to the electric field and its spatial derivative upon approaching the center of the ring from above and below?

Q3. Electrons on a hexagon. [4 marks]

Electrons are placed at six corners of a hexagon inscribed in a circle of radius $R = 1$m, and a proton is placed at the center.
(i) What is the electrical force on the proton? Justify your answer appropriately.
(ii) If the electron in the northwest corner is removed, what is the total electrical force (magnitude & direction) on the proton?

Q4. Spring constants and small oscillations. [4 marks]

Two spheres of mass m and negligible size are connected to the walls nearest to them by identical springs of force constant K. The equilibrium separation between them is r.
(i) When charged to Q Coulombs each, the separation doubles. Obtain K in terms of Q, r & ϵ_0?
(ii) If the left and right spheres are charged to Q and $-Q$ respectively, the separation goes to $r/2$. Once again, obtain K in terms of Q, R & ϵ_0?
(iii) In the situation described in (i) above, supposed that the charge on the right is held fixed while that on the left is displaced by a small amount x and released. Find the effective spring constant k_q of the restoring force $F = -k_q x$ on the left sphere, and the time period T of its oscillations.

Q5. Playing with a dipole. [4 marks]

An isosceles right triangle has its hypotenuse along the x-axis and the apex at $y > 0$. There are charges q and Q placed at the left and right ends of the hypotenuse respectively.
(i) Find the direction of the electric field $\vec{E}$ at the apex as Q is varied in the range $-q \leq Q \leq q$.
(ii) Find the Q for which $|\vec{E}|$ is minimum, and what is its direction?

Indian Institute of Science Education and Research Kolkata

Department of Physical Sciences

PH2201 : Basic Quantum Mechanics
Mid-semester Examination (Spring 2024)
Marks: 20, Duration: 90 minutes

1. In a crude photo-electric experiment, maximum kinetic energy of the emitted electrons are found to be 0.81 eV and 0.40 eV when corresponding incident lights are violet and blue having wavelengths 400 nm and 490 nm respectively. Determine the value of the Planck constant. Will the material show any photo-electric emission if red light having wavelength 620 nm is used? Given $c = 3 \times 10^8$ m s^{-1}. [2+2]

2. Show that the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$ is satisfied by the following actions
$$\hat{x}\psi = -\frac{\hbar}{i}\frac{d\psi}{dp} + \frac{p^2}{2a}\psi \ , \ \hat{p}\psi = p\psi \ ,$$
of the operator $\hat{x}$ and $\hat{p}$ where a is a constant. [3]

3. Using canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$, compute the commutator bracket $[\hat{b}_-\hat{b}_+, \hat{b}_+]$ where $\hat{b}_\pm = \hat{x} \pm i\hat{p}$. [3]

4. (a) Consider a particle of mass m which is moving in a potential given by
$$V(x) = \begin{cases} \infty & x < 0 \\ 0 & 0 \leq x \leq b \\ V_0 & x > b \ . \end{cases}$$
If the energy of the particle is less than V_0 then find the time-independent energy eigenstates up to an undetermined normalization constant.

   (b) Determine the corresponding energy eigenvalues if the constant V_0 is very large. [6+4]

G. M. Hossain

PH2202 Thermal Physics
Fall Semester - 2024
Indian Institute of Science Education and Research, Kolkata
Instructor: Koushik Dutta

Mid-Semester Examination Date: 13/02/2024

No clarification regarding the questions will be provided. If any genuine issue is found in any problem, marks will be given appropriately to all.

1. A process goes through an irreversible change. Is the First Law of Thermodynamics applicable for this change? Choose the correct option among these, and argue briefly.

   (a) Yes (b) No (c) Need to know whether the process completes a full cycle (d) Need to know about the entropy change. **Marks: 1.5**

2. Evaluate the following integral

$$\int_{x=0,y=0,z=0}^{x=1,y=1,z=1} y^2 z dx + 2xyz dy + xy^2 dz \tag{1}$$

   along the path $x^2 + y^2 + z^2 = 5$. **Marks: 1.5**

3. Without introducing the concept of entropy, explain clearly why Joule free expansion is an irreversible process. **Marks: 2**

4. Consider a Carnot cycle for an ideal gas, and draw the cycle in the $p - V$ diagram (p being the y-axis, and V being the x-axis), and mark corners with A, B, C and D, and clearly show isotherms, adiabats and heat flows.

   Now, draw the same process in (i) $T-p$, (ii) $T-V$, (iii) $U-V$, and (iv) $T-S$ diagrams, where T is temperature, p is pressure, V is volume, U is internal energy, and S is entropy. For all the diagrams the first letter should be y-axis, and the second one should be x-axis, e.g for the diagram in (i), y-axis is T and x-axis is p. Also, for each diagram, mark again A, B, C and D clearly. **Marks: 5**

5. Consider a van der Waals' gas for which it is given that $(\partial U/\partial V)_T = a/V^2$ with a being a positive constant. If the gas is allowed to do adiabatic free expansion, show that the temperature will fall as expansion happens. **Marks: 5**

6. For a Carnot engine, the hot reservoir is at temperature T_1, and the cold reservoir is at temperature T_2. To increase its efficiency which one is more effective, increase T_1 (for fixed T_2) or lowering T_2 (for fixed T_1)? Argue well. **Marks: 3**

7. The internal energy of a thermodynamic system is given by $U = Ap^2V$, where A is a constant with appropriate dimension. Determine the equation of the adiabats in the $p - V$ plane. **Marks: 5**

8. Consider a reversible thermodynamic process in which an ideal gas is changed from (p_1, V_1) to (p_2, V_2) where $V_2 > V_1$. In the p-V diagram, the initial and the final state are connected by a straight line. Show that the amount of heat transfer in this process is given by

$$Q = \frac{\gamma}{\gamma - 1}\left[(p_1 + mV_1)(V_2 - V_1) - \frac{m}{2}(1 + \frac{1}{\gamma})(V_2^2 - V_1^2)\right] , \tag{2}$$

   where the straight line has a slope $\frac{dp}{dV} = -m (m > 0)$, and $\gamma = C_p/C_V$. **Marks: 7**

K. Dutta

PH3201 Mid Semester Examination

17 February 2024 3:00-4:30PM

Attempt all questions.

1. Consider a chemical reaction $X \underset{k_1}{\overset{k_2}{\rightleftharpoons}} A$

and assume that the concentration of [A] = *a* is fixed and the number of X reactants are random variable n. The degradation and generation rates are $W_d = k_2\, n$ and $W_g = k_1\, a$

Write the master equation, solve for the steady state P(n), determine mean(n) and variance(n).

4

2. Using Liouville's theorem prove that the thermodynamic average of a function *A*(*p*,*q*) obeys

$$\frac{\partial < A >}{\partial t} = < \{A, H\} >$$ where {,} denotes the Poisson bracket w.r.t. *p* and *q*.

4

3. Suppose there are two independent random variables (y_1, y_2) drawn from a probability distribution function $p(x) = \frac{1}{\sigma\sqrt{2\pi}} exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$ with (μ_1, σ_1) and (μ_2, σ_2) respectively. Determine the characteristic function of the random variable $(y_1 - y_2)$. Hence find mean$(y_1 - y_2)$ and variance$(y_1 - y_2)$.

4

4. Find the leading contribution of the integral $\int_{-1}^{+1} \frac{(\cos x)^n}{4+x^2} dx$ for $n \gg 1$.

4

5. Calculate the moment generating function and the characteristic function for a random variable $x \in (-\infty, +\infty)$ given by the probability distribution function

$$f(x) = \frac{1}{\pi(1+x^2)}$$

Is it possible to calculate mean and variance ?

4

DPS, IISER Kolkata

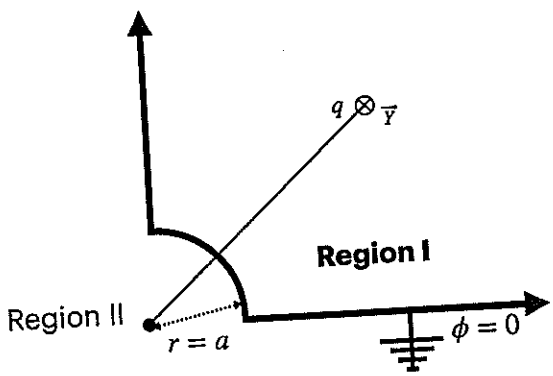
PH3202, Spring 2024

Mid-Semester Test

Instructor: Dr. Rajesh Kumble Nayak
3:00 PM, 14 Feb 2024.
Duration 90 Min.

- Answer all questions.
- Write neatly, and keep the rough work on the answer script. No writing on the question paper.
- Read the question carefully.
- Good luck!

Q - 1(20 Marks):



An infinitely long plate is arrange as shown in the figure. A point charge q is placed in region I, as shown in the figure, at an arbitrary position $\vec{Y}$.

1. Find the potential in the region above the plate, using the method of imaging.
2. Give one point justification for each of the image charges.

Q - 2 (20 Marks):
The following partial differential equation (PDE) for ψ is in three variables, $\{\xi_1, \xi_2, \xi_3\}$,

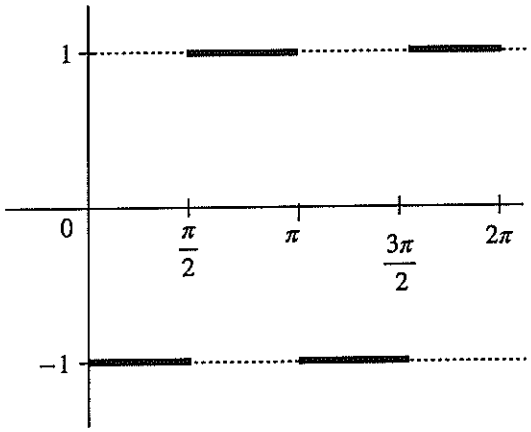
$$\frac{1}{\xi_1}\frac{\partial^2}{\partial {\xi_1}^2}\left(\xi_1\psi\right) + \frac{1}{{\xi_1}^2 \sin\xi_2}\frac{\partial}{\partial\xi_2}\left(\sin\xi_2\frac{\partial\psi}{\partial\xi_2}\right) + \frac{1}{{\xi_1}^2\sin^2\xi_2}\frac{\partial^2}{\partial {\xi_3}^2}\left(\psi\right) = 0\,.$$

If the PDE is separable, write it as three ordinary differential equations in appropriate variables.

Q - 3 (20 Marks):
Express the following function in terms of Fourier series.

$$f(x) = \begin{cases} -1 & 0 < \theta < \frac{\pi}{2} \\ 1 & \frac{\pi}{2} < \theta < \pi \\ -1 & \pi < \theta < 3\frac{\pi}{2} \\ 1 & 3\frac{\pi}{2} < \theta < 2\pi \end{cases},$$

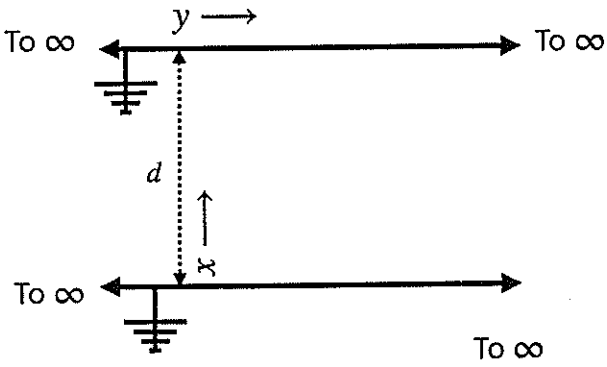
A plot of the function is given below:



Q - 4 (20 Marks):

Two infinity long ground parallel metallic plates, separated by a distance d are arranged as shown in the figure. Ignoring the y dependence of the problem,

1. Find the electrostatic potential, ϕ, in the region between the two plates.
2. Find the electrostatic field, $\vec{E}$, in the region between the two plates.
3. If a point test charge, q, is placed near the metallic wall, in which direction it will move.



Hint! Is it enough to treat the system as 1-D or we need to solve full 3-D problem.

PH3203: Advanced Quantum Mechanics

Mid Semester Examination

19th February 2024

Total Marks: 30 **Time - 1 hour 30 minutes**

Instructions: *Mark the correct option/options in the question paper. Full credit will be given only if all the correct options are marked. Marking any incorrect option will lead to zero credit.*

Name..... **Roll No.**.....

PART - I

Marks: (2 × 5 = 10)

Q.1) Which one of the following operators is not a linear operator?

(A) $\hat{O}\psi(x) = \frac{d\psi(x)}{dx}$

(B) $\hat{O}\psi(x) = -\psi(x)$

(C) $\hat{O}\psi(x) = \psi^*(x)$

(D) $\hat{O}\psi(x) = \psi(x + x_0)$, with x_0 being a constant

Q.2) The Hamiltonian for a particle is changed over a time period τ. The minimum gap in the spectrum of the initial Hamiltonian is ΔE_{min}. Which one of the following conditions defines an adiabatic approximation?

(A) $\tau >> \frac{\hbar}{\Delta E_{min}}$

(B) $\tau << \frac{\hbar}{\Delta E_{min}}$

(C) $\tau \lesssim \frac{\hbar}{\Delta E_{min}}$

(D) $\tau \gtrsim \frac{\hbar}{\Delta E_{min}}$

Q.3) For which of the following density matrices ($\hat{\rho}$), the condition $\hat{\rho}^2 = \hat{\rho}$ does not hold?

Sourin Das

(A) $\hat{\rho} = \frac{1}{2}(\mathbb{I} + \sigma_x)$

(B) $\hat{\rho} = \frac{1}{2}\left(\mathbb{I} + \frac{1}{\sqrt{2}}\sigma_x + \frac{1}{\sqrt{2}}\sigma_z\right)$

(C) $\hat{\rho} = \frac{1}{2}\left(\mathbb{I} + \cos(\frac{\theta}{2})e^{i\phi}\sigma_x + \sin(\frac{\theta}{2})e^{-i\phi}\sigma_y\right)$

(D) $\hat{\rho} = \frac{1}{2}\left(\mathbb{I} + \frac{1}{2}\sigma_y + \frac{1}{2}\sigma_z\right)$

Note: $\mathbb{I}$ is the 2×2 identity matrix and $\sigma_x, \sigma_y, \sigma_z$ are Pauli spin-matrices for spin-$\frac{1}{2}$ particles.

Q.4) The eigenvalues of the operator $\vec{S} \cdot \hat{n}$ for a spin-1 particle,

(A) $-\frac{\hbar}{2}, 0, \frac{\hbar}{2}$

(B) $-\frac{3\hbar}{2}, 0, \frac{3\hbar}{2}$

(C) $-\hbar, 0, \hbar$

(D) $-2\hbar, 0, 2\hbar$

Note: $\vec{S} = (S_x, S_y, S_z)$ and $\hat{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ is an arbitrary unit vector in 3-dimension. The matrix representations are: $S_x = \frac{\hbar}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, S_y = \frac{\hbar}{\sqrt{2}}\begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, S_z = \hbar\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.

Q.5) Consider a spin-$\frac{1}{2}$ particle subjected to a magnetic field along the z-direction ($B\hat{z}$). The dynamical phase accumulated by the particle when it is initially ($t = 0$) prepared at one of the eigenstates of σ_x, i.e, $|+x\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$, where $|\uparrow\rangle$ and $|\downarrow\rangle$ are eigenstates of σ_z matrix with eigenvalues $+1$ and -1 respectively, is

(A) $\frac{\pi}{2}$

(B) π

(C) 2π

(D) None of the above

PART - II

Marks: (10+10 = 20)

Q.6) (a) The density operator for a pure state of a spin-$\frac{1}{2}$ particle with its polarization point-

ing along an arbitrary direction $\hat{n}$ is given by $\hat{\rho} = \frac{1}{2}(\mathbb{I} + \hat{n} \cdot \vec{\sigma})$. For a Hamiltonian $\hat{H} = \alpha\sigma_z$ (α has the dimension of energy, $\hbar = 1$), using Heisenberg equations of motion, the periodic time evolution of $\hat{n}$ is given by,

(A) $\hat{n}(t) = \Big(A\cos(\alpha t) + B\sin(\alpha t)\Big)\hat{i} + \Big(A\sin(\alpha t) - B\cos(\alpha t)\Big)\hat{j} + C\hat{k}$

(B) $\hat{n}(t) = \Big(A\cos(\alpha t) + B\sin(\alpha t)\Big)\hat{i} + \Big(A\sin(\alpha t) - B\cos(\alpha t)\Big)\hat{j}$

(C) $\hat{n}(t) = \Big(A\cos(2\alpha t) + B\sin(2\alpha t)\Big)\hat{i} + \Big(A\sin(2\alpha t) - B\cos(2\alpha t)\Big)\hat{j} + C\hat{k}$

(D) $\hat{n}(t) = \Big(A\cos(\frac{\alpha}{2}t) + B\sin(\frac{\alpha}{2}t)\Big)\hat{i} + \Big(A\sin(\frac{\alpha}{2}t) - B\cos(\frac{\alpha}{2}t)\Big)\hat{j} + C\hat{k}$

Note: Here A, B, C are constants with $|A|^2 + |B|^2 + |C|^2 = 1$.

(b) Using the fact that $e^{i\hat{H}t}$ represents a time evolution operator, the wave function for a spin-$\frac{1}{2}$ system after one period of evolution becomes (assume the initial state is $|+x\rangle = \frac{1}{\sqrt{2}}[1 \quad 1]^T$)

(A) $\frac{1}{\sqrt{2}}[1 \quad -1]^T$

(B) $\frac{1}{\sqrt{2}}[1 \quad -i]^T$

(C) $\frac{1}{\sqrt{2}}[-1 \quad -1]^T$

(D) $\frac{1}{\sqrt{2}}[1 \quad 1]^T$

(c) The geometric phase accumulated by the wavefunction for an arbitrary initial state $|\psi\rangle = \cos(\theta/2)|\uparrow\rangle + e^{i\phi}\sin(\theta/2)|\downarrow\rangle$ over one period of evolution is given by,

(A) Ω

(B) $\frac{1}{2}\Omega$

(C) 2Ω

(D) $\frac{1}{4}\Omega$

Note: Here Ω is the solid angle subtended at the origin by the closed-loop trajectory of the evolved state vector tip. $|\uparrow\rangle$ and $|\downarrow\rangle$ are eigenstates of σ_z matrix with eigenvalues $+1$ and -1 respectively.

(d) Now if we consider a spin-1 particle, initially prepared in the eigenstate of $\hat{S}_x$, i.e., $\frac{1}{2}[1 \quad \sqrt{2} \quad 1]^T$, for a Hamiltonian $\hat{H} = \alpha\hat{S}_z$, the geometric phase accumulated over one period of evolution is given by,

(A) $\frac{\pi}{2}$

(B) π

(C) 2π

(D) 4π

Hint: The arbitrary spin-1 particle state is given by $|\Psi\rangle = \begin{pmatrix} e^{-i\phi}\cos^2(\frac{\theta}{2}) \\ \sqrt{2}\cos(\frac{\theta}{2})\sin(\frac{\theta}{2}) \\ e^{i\phi}\sin^2(\frac{\theta}{2}) \end{pmatrix}$. The matrix representations are: $S_x = \frac{\hbar}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, S_y = \frac{\hbar}{\sqrt{2}}\begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, S_z = \hbar\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.

Q.7) Consider a particle of mass m and charge q is initially ($t < 0$) trapped in the ground state of a harmonic oscillator potential $\frac{1}{2}m\omega^2\hat{x}^2$. At $t = 0$, a constant electric field $\mathcal{E}$ is suddenly switched on.

Note: If the ground state for $t < 0$ is $|0\rangle$, then in terms of the new eigenstates at $t > 0$ i.e., $|\tilde{n}\rangle$, $|0\rangle = \sum_{n=0}^{\infty} \tilde{C}_n|\tilde{n}\rangle$, where $\sum_{n=0}^{\infty} |\tilde{C}_n|^2 = 1$.

(a) The evolution of the position operator is given by $\hat{x}(t)$. Which one of the following statements is true ?

(A) $\langle\hat{x}(t > 0)\rangle = \sqrt{\frac{\hbar}{2m\omega}}\sum_{n=0}^{\infty}\sqrt{n+1}(\tilde{C}_n^*\tilde{C}_{n+1}e^{-in\omega t} + \tilde{C}_{n+1}^*\tilde{C}_n e^{in\omega t})$

(B) $\langle\hat{x}(t > 0)\rangle = -\frac{q\mathcal{E}}{m\omega^2} + \sqrt{\frac{\hbar}{2m\omega}}\sum_{n=0}^{\infty}\sqrt{n+1}(\tilde{C}_n^*\tilde{C}_{n+1}e^{-in\omega t} + \tilde{C}_{n+1}^*\tilde{C}_n e^{in\omega t})$

(C) $\sqrt{\frac{\hbar}{2m\omega}}\sum_{n=0}^{\infty}\sqrt{n+1}(\tilde{C}_n^*\tilde{C}_{n+1} + \tilde{C}_{n+1}^*\tilde{C}_n) = \frac{q\mathcal{E}}{m\omega^2}$

(D) $\sqrt{\frac{\hbar}{2m\omega}}\sum_{n=0}^{\infty}\sqrt{n+1}(\tilde{C}_n^*\tilde{C}_{n+1} + \tilde{C}_{n+1}^*\tilde{C}_n) = -\frac{q\mathcal{E}}{m\omega^2}$

(b) The evolution of the momentum operator is given by $\hat{p}(t)$. Which one of the following statements is true ?

(A) $\langle\hat{p}(t > 0)\rangle = \frac{q\mathcal{E}}{\omega} - i\sqrt{\frac{m\omega\hbar}{2}}\sum_{n=0}^{\infty}\sqrt{n+1}(\tilde{C}_n^*\tilde{C}_{n+1}e^{-i\omega t} - \tilde{C}_{n+1}^*\tilde{C}_n e^{i\omega t})$

(B) $\langle\hat{p}(t > 0)\rangle = -i\sqrt{\frac{m\omega\hbar}{2}}\sum_{n=0}^{\infty}\sqrt{n+1}(\tilde{C}_n^*\tilde{C}_{n+1}e^{-in\omega t} - \tilde{C}_{n+1}^*\tilde{C}_n e^{in\omega t})$

(C) $-i\sqrt{\frac{m\omega\hbar}{2}}\sum_{n=0}^{\infty}\sqrt{n+1}(\tilde{C}_n^*\tilde{C}_{n+1} - \tilde{C}_{n+1}^*\tilde{C}_n) = -\frac{q\mathcal{E}}{\omega}$

(D) $-i\sqrt{\frac{m\omega\hbar}{2}}\sum_{n=0}^{\infty}\sqrt{n+1}(\tilde{C}_n^*\tilde{C}_{n+1} - \tilde{C}_{n+1}^*\tilde{C}_n) = 0$

Useful Information:

$$\vec{\nabla}f = \frac{\partial f}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial f}{\partial\theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial f}{\partial\phi}\hat{\phi} \tag{1}$$

$$\vec{\nabla}\times\vec{A} = \frac{1}{r\sin\theta}\Big(\frac{\partial}{\partial\theta}(A_\phi\sin\theta) - \frac{\partial A_\theta}{\partial\phi}\Big)\hat{r} + \frac{1}{r}\Big(\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{\partial}{\partial r}(rA_\phi)\Big)\hat{\theta} + \frac{1}{r}\Big(\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial\theta}\Big)\hat{\phi} \tag{2}$$

Indian Institute of Science Education and Research, Kolkata
Department of Physical Sciences
Advanced Stat Physics (PH4202)
Instructor: *P. K. Mohanty*

Midsem Exam (15th Feb, 2024, 10:00-11:30) **Full Mark: [20]**

Attempt Question 1 (Q1) and any two *of the rest (Q2, Q3, Q4, Q5)*

Notations: $\beta = \frac{1}{k_B T}$, T is temperature and k_B is Boltzman constant.

1. Consider the diluted Ising model (also called Blume Campel model) on a periodic one dimensional lattice of size L, where each site $i = 1, 2, \ldots, L$ with a site variable $s_i = -1, 0, 1$ describing an empty site ($s_i = 0$ or a site occupied by a particle with spin +1 ($s_i = 1$) or -1 ($s_i = -1$). Energy of a configuration $\{s_i\}$ is
$$E(\{s_i\}) = -\sum_{i=1}^{L} s_i s_{i+1} + \mu \sum_{i=1}^{L} s_i^2 - h \sum_{i=1}^{L} s_i \tag{1}$$
where chemical potential μ controls the number of particles $N = \sum_{i=1}^{L} s_i^2$ and magnetic filed controls the magnetization $M = \sum_{i=1}^{L} s_i$ of the system.
(a) Use meanfiled theory, where s_i fluctuates a little from its mean value $\langle s_i \rangle = m$ where $M = Lm$, and express the free energy persite of the syste as
$$f = a_0 + a_2 m^2 + a_4 m^4 \tag{2}$$
when $h = 0$. Express a_0, a_2, a_4 in terms of $k = e^{-\beta\mu}$ and β. **Hint:** Keep μ-term as it is. [5]
(b) Such an expansion is valid when $a_4 > 0$, else you must consider next higher order term. Find the regime of k for which Eq. (2) is valid. [1]
(c) Find the phase diagram in k-β plane that separates the ordered (ferromagnetic) phase from the disordered (paramagnetic phase). [1]
(d) Re-draw the phase diagram in $\mu - \beta$ phase-plane. [1]
(e) What additional term do you expect in Eq. (2) for a small field h ? [1]
(f)Do you think that the critical exponent $\nu, \beta, \gamma, \delta$ will differ from a model that has Ising or Z(2) symmetry ? Explain why, or why not ? [1]

2. For the model defined in Eq. (1), express the exact partition function $Z(\beta, \mu, h)$ as $Tr(M^L)$.
(a) Find the transfer matrix M. [2]
(b)Assume that the largest eigenvalue of M is λ. Express the mean density of the system $\rho = \frac{\langle N \rangle}{L}$ and the compressibility $\frac{1}{L}(\langle N^2 \rangle - \langle N \rangle^2)$ of the system in terms λ and its derivatives. [1.5+1.5]

3. Near the critical point the order parameter ϕ and the sucpeptibility χ (with respective critical exponents β and γ) of a system are given by
$$\phi(\epsilon) = \int_0^\infty q(x, \epsilon) dx; \quad \chi(\epsilon) = \int_0^\infty x^2 q(x, \epsilon) dx \tag{3}$$
where the function $q(x, \epsilon)$ has a scaling form $q(x, \epsilon) = x^{-a} g(\frac{\epsilon}{x^b})$. Express the critical exponents a and b in terms of β and γ. [5]

4. Consider Ising model on a periodic one dimensional lattice of size L, where each site $i = 1, 2, \ldots, L$ carry a spin variable $s_i = \pm 1$. Energy of a configuration $\{s_i\}$ is $E(\{s_i\}) = -\sum_{i=1}^{L} s_i s_{i+1}$. If the spins at site k and $k + 2r$ with $r < \frac{L}{4}$ are frozen to a value +1, how does the magnetization at the site $k + r$ vary with temperature T ? [5]

5. XY model: On a periodic one dimensional lattice two-component unit vectors $\vec{s}_i$ with $i = 1, 2, \ldots, L$ interact as follows:
$$E(\{\vec{s}_i\}) = -\sum_{i=1}^{L} \vec{s}_i . \vec{s}_{i+1} = -\sum_{i=1}^{L} \cos(\theta_i - \theta_{i+1}), \tag{4}$$
where θ_i is the angle that $\vec{s}_i$ makes with a fixed axis. Express the partition function of the system at temperature T in terms of modified Bessel function $I_0(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{x \cos(\theta)} d\theta$. [5]

15/02/2024

Indian Institute of Science Education and Research, Kolkata
Department of Physical Sciences
Advanced Stat Physics (PH4202)
Instructor: *P. K. Mohanty*

IISER KOLKATA

Midsem Exam (15th Feb, 2024, 10:00-11:30) **Full Mark: [20]**

Attempt Question 1 (Q1) and any two *of the rest (Q2, Q3, Q4, Q5)*

Notations: $\beta = \frac{1}{k_B T}$, T is temperature and k_B is Boltzman constant.

1. Consider the diluted Ising model (also called Blume Campel model) on a periodic one dimensional lattice of size L, where each site $i = 1, 2, \ldots, L$ with a site variable $s_i = -1, 0, 1$ describing an empty site ($s_i = 0$ or a site occupied by a particle with spin +1 ($s_i = 1$) or -1 ($s_i = -1$). Energy of a configuration $\{s_i\}$ is
$$E(\{s_i\}) = -\sum_{i=1}^{L} s_i s_{i+1} + \mu \sum_{i=1}^{L} s_i^2 - h \sum_{i=1}^{L} s_i \tag{1}$$
where chemical potential μ controls the number of particles $N = \sum_{i=1}^{L} s_i^2$ and magnetic filed controls the magnetization $M = \sum_{i=1}^{L} s_i$ of the system.
(a) Use meanfiled theory, where s_i fluctuates a little from its mean value $\langle s_i \rangle = m$ where $M = Lm$, and express the free energy persite of the syste as
$$f = a_0 + a_2 m^2 + a_4 m^4 \tag{2}$$
when $h = 0$. Express a_0, a_2, a_4 in terms of $k = e^{-\beta\mu}$ and β. **Hint:** Keep μ-term as it is. [5]
(b) Such an expansion is valid when $a_4 > 0$, else you must consider next higher order term. Find the regime of k for which Eq. (2) is valid. [1]
(c) Find the phase diagram in k-β plane that separates the ordered (ferromagnetic) phase from the disordered (paramagnetic phase). [1]
(d) Re-draw the phase diagram in $\mu - \beta$ phase-plane. [1]
(e) What additional term do you expect in Eq. (2) for a small field h ? [1]
(f)Do you think that the critical exponent $\nu, \beta, \gamma, \delta$ will differ from a model that has Ising or Z(2) symmetry ? Explain why, or why not ? [1]

2. For the model defined in Eq. (1), express the exact partition function $Z(\beta, \mu, h)$ as $Tr(M^L)$.
(a) Find the transfer matrix M. [2]
(b)Assume that the largest eigenvalue of M is λ. Express the mean density of the system $\rho = \frac{\langle N \rangle}{L}$ and the compressibility $\frac{1}{L}(\langle N^2 \rangle - \langle N \rangle^2)$ of the system in terms λ and its derivatives. [1.5+1.5]

3. Near the critical point the order parameter ϕ and the sucpeptibility χ (with respective critical exponents β and γ) of a system are given by
$$\phi(\epsilon) = \int_0^\infty q(x, \epsilon) dx; \quad \chi(\epsilon) = \int_0^\infty x^2 q(x, \epsilon) dx \tag{3}$$
where the function $q(x, \epsilon)$ has a scaling form $q(x, \epsilon) = x^{-a} g(\frac{\epsilon}{x^b})$. Express the critical exponents a and b in terms of β and γ. [5]

4. Consider Ising model on a periodic one dimensional lattice of size L, where each site $i = 1, 2, \ldots, L$ carry a spin variable $s_i = \pm 1$. Energy of a configuration $\{s_i\}$ is $E(\{s_i\}) = -\sum_{i=1}^{L} s_i s_{i+1}$. If the spins at site k and $k + 2r$ with $r < \frac{L}{4}$ are frozen to a value +1, how does the magnetization at the site $k + r$ vary with temperature T ? [5]

5. XY model: On a periodic one dimensional lattice two-component unit vectors $\vec{s}_i$ with $i = 1, 2, \ldots, L$ interact as follows:
$$E(\{\vec{s}_i\}) = -\sum_{i=1}^{L} \vec{s}_i . \vec{s}_{i+1} = -\sum_{i=1}^{L} \cos(\theta_i - \theta_{i+1}), \tag{4}$$
where θ_i is the angle that $\vec{s}_i$ makes with a fixed axis. Express the partition function of the system at temperature T in terms of modified Bessel function $I_0(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{x \cos(\theta)} d\theta$. [5]

15/02/2024

Midd-semester examination: PH4204 (2024)

Time : 1 hours 30 minutes : Full marks = 25

Answer all questions.

1. Write down the quark composition of each of the following hadrons:
   (a) π^+, (b) ρ^+, (c) K^-, (d) $\bar{p}$, (e) $\bar{K}^0$.

[5 × 1]

2. Is it possible to have the following decays, driven by strong interaction:
   (a) $\rho^0 \to \pi^+\pi^-$, (b) $\rho^0 \to \pi^0\pi^0$?
   In each case, given reasons after checking all relevant conservation principles.
   $(J(\rho^0) = 1, P(\rho^0) = -1, I(\rho) = 1)$

[3+3]

3. Assuming that Maxwell's equations on electrodynamics are invariant under parity, show that $\vec{E}$, the electric field vector, is odd under parity.

[6]

4. Consider the two-body decay $A \to BC$. Suppose $m_A = 100$ GeV, $m_B = 40$ GeV, and $m_C = 20$ GeV.
   Now, starting from the principles of conservation of energy and momentum, obtain numerically the energies of B and C in the rest frame of A.
   (c = 1).

[4+4]

IISER Kolkata
Mid-Semester Examination, 2024: Fourth Year, Semester II;
PH4205 (General Theory of Relativity and Cosmology)
Full Marks 20
Answer all questions
Time: 1 hour 30 minutes

1. Show that a vector orthogonal to a timelike vector must be spacelike. You can use the Minkowski metric.

2. A 2-d metric is given as $ds^2 = dV^2 - V^2 dU^2$. Find out all the non-zero Chrisoffel symbols.

3. Given that ϕ is a scalar, show that the covariant divergence $(g^{\mu\nu}\phi_{,\nu})_{;\mu}$ can be written as an ordinary divergence.

4. Starting from the Bianchi identity, $R^{\alpha}_{\beta(\gamma\delta;\lambda)} = 0$, arrive at its contracted form $(R^{\alpha\beta} - \frac{1}{2}Rg^{\alpha\beta})_{;\beta} = 0$.

5. Consider the Minkowski metric $ds^2 = c^2dt^2 - dx^2 - dy^2 - dz^2$.

i)Write down the metric in terms of the coordinates $\psi = ct + x$, $\phi = ct - x$ while y, z remain the same.

ii)Check that the signature requirement is satisfied.

iii) Comment on the nature of the coordinate ϕ.

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Useful formula (if required!):
1. $\Gamma^{\mu}_{\alpha\beta} = \frac{1}{2}g^{\mu\nu}(g_{\alpha\nu,\beta} + g_{\nu\beta,\alpha} - g_{\alpha\beta,\nu})$

PH 4207, Mid Sem Exam (Total Marks 20)

(13/02/24) $e^{i\phi}$

1. (a) Consider the Bloch state on the Bloch sphere, given by $|\psi\rangle = cos\frac{\theta}{2}|0\rangle + exp(-i\phi)sin\frac{\theta}{2}|1\rangle$.
Show explicitly using unitary rotation on the Bloch sphere, how can one go from the state $|0\rangle$ in the Bloch sphere to $|\psi\rangle$ in **exactly one** step. Write down explicitly the axis about which this rotation is to be performed. [5]

   (b) In the above example, please also show using Euler decomposition, how one can reach $|\psi\rangle$ from $|0\rangle$ on the Bloch sphere. You can use a maximum of up to 3 rotations/decomposition. [3]

   (c) Starting from a general 2 x 2 matrix of complex entries,
$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$;
where a,b,c and d are elements of C, find out the number of free parameters if g $\in$ SU(2). Prove your results explicitly. [3]

2. Consider the singlet state comprising of a two particle state,
$|\psi\rangle = \frac{1}{\sqrt{2}}[|01\rangle - |10\rangle]$, where $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
Show that the quantum mechanical expectation value,

$$E(n,m) = \langle\psi|\vec{\sigma}.\hat{n} \otimes \vec{\sigma}.\hat{m}|\psi\rangle,$$

is given by $E(n,m) = -cos\phi_{n,m}$, where, $\phi_{n,m}$ is the angle between unit vectors $\hat{m}$ and $\hat{n}$. [4]

3. Consider the three qubit W state

$$|\psi\rangle = \frac{1}{\sqrt{2}}[|001\rangle + |010\rangle + |100\rangle]$$

$\sqrt{3}$

   (a) Find out the reduced density matrix $\rho_{AB} = Tr_C(\rho_{ABC})$ [1.5]
   (b) Calculate the Von-Neumann entropy , S(ρ_{AB}) [1]
   (c) Find out $\rho_A = Tr_{BC}(\rho_{ABC})$. Calculate S($\rho_A$) [1.5]
   (d) Comment on the nature of entanglement of the W state. [1]

Date : 12th Feb, 2024 Mid-semester Exam

PH4208 : Quantum Field Theory 2

Duration: 90 minutes Weight: 20 %

Important Identites

- 2-dim $SU(2)$ matrices:

$$T^1 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad T^2 = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad T^3 = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- 3-dim $SU(2)$ matrices:

$$T^1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \qquad T^2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \qquad T^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

1. **Dirac equation in $2+1$ dim:** Consider $2+1$ space-time and the Dirac equation in terms of γ^μ matrices.

   (a) List the properties of γ^μ matrices that are valid in $2+1$ dimenstions. 2

   (b) What is the minimum possible size of γ^μ matrices in $2+1$ dimensions? Write a possible choice (for minimum size) of γ^μ matrices and show that they satisfy the properties in previous part. 4

2. $SU(2)$ **gauge theory:** Consider a two component Dirac fermion Ψ and a three component scalar Φ given as:

$$\Psi = \begin{pmatrix} p \\ n \end{pmatrix} \qquad \text{and} \qquad \Phi = \begin{pmatrix} \pi^+ \\ \pi^0 \\ \pi^- \end{pmatrix}$$

   Under $SU(2)$ gauge transformation we have $\Psi(x) \to U(x)\Psi(x)$ and $\Phi(x) \to V(x)\Phi(x)$, where $U(x)$ is 2×2 and $V(x)$ is 3×3 unitary matrix and elements of $SU(2)$ group.

   (a) Write down the Lagrangian density for Ψ (with mass m_1) and Φ (with mass m_2) and expand in terms of the gauge fields. 3

   (b) Using the explicit form of T^a matrices for 2-dim and 3-dim cases, expand the above Lagrangian completely in terms of component of fields Ψ, Φ and A^a_μ. 8

   (c) Consider the Yukawa term $\bar{\Psi}\sigma^a\gamma^5\Psi\Phi^a$, where $\sigma^a = \{\sigma^+, \sigma^z, \sigma^-\}$ are three Pauli matrices with $\sigma^\pm = (\sigma^x \pm i\sigma^y)/2$. Expand this term in terms of component fields. 3

   (d) **Bonus question:** What can you say about the transformation properties of the Yukawa term under $SU(2)$ transformation given above. Ungraded

Evolutionary Dynamics (PH4209) Mid-Sem exam (Feb.17, 2024)

Answer all questions. Total Time=1.5 hours. Total Marks=20.The mark for each question is specified in brackets.

1. a) For a population consisting of two types of strategies A and B whose frequency dependent fitness's are f_A and f_B respectively, write down the quasi-species equation (assuming no mutation) showing the rate of change of frequency of type A. If a<c and d<b, obtain the possible equilibrium frequencies for A. a,b,c,d correspond to the matrix elements E(A,A), E(A,B), E(B,A), E(B,B) respectively of the payoff matrix. (2)

b) Using the flow diagram technique, prove the equilibrium solution obtained in a) is a *stable* equilibrium. (2)

c) In a Hawk (H), Dove (D) and Bourgeoise (B) game, B acts like H when it arrives at a resource first and acts like D when it arrives at the resource after the opponent. Write down the payoff matrix for such a game, the fitness of each strategy and obtain the condition on the parameters a,b,c,d that needs to be satisfied if a small fraction of B is to invade a population consisting a mixture of H and D players at equilibrium. (3)

2. A finite population of size N is made up of two types of individuals A and B whose initial numbers in the population are **i** and **N-i** respectively. The population undergoes evolution by the Moran process.

a) If there is no difference between fitness between types A and B, what is the probability of choosing A for death and B for reproduction? (1)

b) If $\mathbf{x_i}$ is the fixation probability of type A, write down the equation of $\mathbf{x_i}$ in terms of $\mathbf{x_{i-1}}$ and $\mathbf{x_{i+1}}$ (1)

c) Solve the above equation to obtain the fixation probability of A starting from an initial state where just ***one*** A-type is present in the population along with N-1 other B-types. (3)

d) If A and B are involved in a game, write down the total payoff to an A-player, a B-player when the state of the population consists of *i* A players and *(N-i)* B-players. (2)

i. What is the probability that A is chosen for reproduction and B for death?
ii. What is the probability that B is chosen for reproduction and A for death?

3. Consider two stochastic strategies with last round memory defined by S1(p1,p2,p3,p4) and S2(q1,q2,q3,q4) where p1,p2,p3,p4 are S1's probabilities of cooperating when the state of the game in the last round was CC, CD, DC, DD respectively.

(a) Write down the matrix giving the transition probabilities between the 4 possible Markov states. (2)
(b) If μ is the mutation probability of making a mistake (behavioural noise); obtain the transition probabilities in going from (i) *CC to CC* (ii) *DC to DD* in the presence of such behavioural noise. Justify each term in your expressions for the transition probabilities (4)

PH4213 Mid-semester Examination

Full Marks 20 **Time : 90 minutes**

Attempt all questions

Q 1) i) If N_1 and N_2 are normal subgroups of a group G with $N_1 \cap N_2 = \{e\}$, where e is the identity of G , then show that every element of N_1 commutes with every element of N_2.

ii) Let $\phi : G_1 \to G_2$ be a homomorphism. Prove that

$$\ker \phi \equiv \{g \in G_1 \, : \, \phi(g) = e_2\}$$

is a subgroup of G_1. $(3+2)$

Q 2) A Lorentz transformation can be represented by a 4×4 matrix Λ that obeys $\Lambda^T \eta \Lambda = \eta$. Show that

(i) $\det \Lambda = \pm 1$.

(ii) $|\Lambda^0{}_0| \geq 1$. *Hint : look at the 00-th element of* $\Lambda^T \eta \Lambda$ $\left(1\frac{1}{2} \times 2\right)$

Q 3) i) Show that under an **active** rotation through an angle θ about an axis determined by the unit vector $\hat{n}$ a vector $\vec{r}$ changes to

$$\vec{r}' = \cos\theta \vec{r} + (1 - \cos\theta)(\hat{n} \cdot \vec{r})\hat{n} + \sin\theta(\hat{n} \times \vec{r})$$

ii) Hence write down the matrix $R(\hat{n}, \theta)$ representing a **passive** rotation through an angle θ about an axis $\hat{n}$. *You may write down the matrix either in a 3×3 array form or simply write down the ij-th element of the matrix.*

iii) Directly verify by matrix multiplication (again, feel free to use either the explicit form of the matrix or an expression for the ij-th element) that $R(\hat{n}, -\theta) = R(-\hat{n}, \theta) = R(\hat{n}, \theta)^{-1}$.

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iv) Prove directly from the matrix (or from the expression for the ij-th element) that

$$\cos\theta = \frac{1}{2}\left(\mathrm{Tr}\ R(\hat{n},\theta) - 1\right)$$

v) Three passive rotations are carried out in succession. The first one is about the Z axis through an angle $\frac{\pi}{4}$. The second is one about the X axis through the angle $\frac{\pi}{2}$. The third one is again about the Z axis, and is through the angle $\frac{\pi}{2}$. Find the net angle of the resulting overall rotation.

vi) What are the Euler angles α, β and γ for the overall rotation in part (iv) above? $(3+1+2+2+3+1)$

Thanks Dr Gopakumar